Sumaiya Rahman

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Sumaiya Rahman | sumaiya | Google Scholar | ORCID

About

I am a Computer Science and Engineering graduate and currently serving as a Lecturer. My academic and research journey has centered on Artificial Intelligence, Machine Learning, and Natural Language Processing, leading to impactful projects and publications in international conferences. These experiences have strengthened my commitment to continuous learning, effective teaching, and knowledge sharing. I am passionate about fostering innovation in both research and education and aim to contribute meaningfully to the academic and scientific community.

Research Interests

- Machine Learning & Deep Learning
- Artificial Intelligence
- Computer Vision
- Natural Language Processing (NLP)

Education

- **Rajshahi University of Engineering & Technology (RUET)** *Oct 2025 – Present*
  M.Sc. in Computer Science and Engineering Rajshahi, Bangladesh
  - **CGPA: 4.00 / 4.00** (1st Semester)
- **Rajshahi University of Engineering & Technology (RUET)** *Jan 2020 – Jun 2025*
  B.Sc. in Computer Science and Engineering Rajshahi, Bangladesh
  - **CGPA: 3.75 / 4.00**
- **Holy Cross College** *Jul 2017 – Jul 2019*
  Higher Secondary Certificate (Science) Tejgaon, Dhaka, Bangladesh
  - **GPA: 5.00 / 5.00** (Dhaka Board)
- **Shaheed Police Smrity School and College** *Jan 2015 – Jun 2017*
  Secondary School Certificate (Science) Mirpur, Dhaka, Bangladesh
  - **GPA: 5.00 / 5.00** (Dhaka Board)

Experience

- **Lecturer, Department of Computer Science and Engineering (CSE)** *Full Time*
  Eastern University, Dhaka, Bangladesh June 2026 – Present
  - Teaching undergraduate courses and laboratory sessions in Computer Science and Engineering.
  - Supervising undergraduate thesis and capstone projects.
  - Developing course materials and assessment components.
  - Providing academic advising and student mentorship.
- **Lecturer, Department of Computer Science and Engineering (CSE)** *Full Time*
  World University of Bangladesh, Dhaka, Bangladesh July 2025 – May 2026
  - Conducted undergraduate lectures and laboratory classes in Computer Science and Engineering.
  - Prepared course materials, assignments, and examinations.
  - Evaluated student performance and academic progress.
  - Assisted in departmental academic and administrative activities.

PUBLICATIONS

C=CONFERENCE

[C.1] Sumaiya Rahman *et al.* (2025). **Multimodal Deep Learning Approach for Bangla Sign Language Recognition: Integrating Spatial and Geometric Features**. In the *2nd International Conference on Next-Generation Computing, IoT, and Machine Learning*, 2025.

[C.2] Shaikh Ibna Sazzad, Sumaiya Rahman *et al.* (2025). **Detecting Depression in Social Media Posts Using SBERT-CNN: A Transformer-based Semantic Neural Network Approach**. In the *28th IEEE International Conference on Computer and Information Technology (ICCIT)*, Bangladesh, 2025.

[C.3] Mayeesha Farjana, Sumaiya Rahman *et al.* (2025). **Classifying Fake News With A Large-Scale Dataset Using Machine Learning Approaches**. In the *IEEE International Conference on Quantum Photonics, Artificial Intelligence, and Networking*, 2025.

[C.4] Md. Yesrif Hasan Emon, Shyla Afroge, Mahmudun Nabi Saikat, Sumaiya Rahman *et al.* (2025). **ER-SE Hybrid Model for Automated Kidney Abnormality Detection in CT Images**. In the *IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON)*, Bangladesh, 2025.

[C.5] Mayeesha Farjana, Barisha Chowdhury, Farhana Rahman, Zuairia Raisa Bintay Makin, Sumaiya Rahman *et al.*(2023). **Gender Abusive Language Detection in Bengali Using Machine Learning Algorithms**. In the *2nd International Conference on Big Data, IoT and Machine Learning*, pp.1-15. Springer, 2023.

MANUSCRIPTS UNDER REVIEW

[J.1] Md Minhazul Islam, Md Tanbeer Jubaer, Sumaiya Rahman *et al.* (2026). **YTClickbait21K: Human-Annotated Multimodal Dataset for YouTube Clickbait Detection Across Diverse Channels and Content Categories**. Submitted to *Scientific Data*, Springer Nature **[Q1 Journal]** .

PROJECTS

- **Bangla Hate Speech Detection** *CSE 3200 Project*
  Tools: Python, BiLSTM, TF-IDF
  ◦ Built a BiLSTM model using a 5K self-collected dataset for detecting Bangla hate comments. Applied TF-IDF, preprocessing, and evaluated with precision & recall.

- **Luminous Learners Portal** *CSE 3100 Project*
  Tools: Django, HTML, CSS, Bootstrap
  ◦ Developed a study portal website for managing lectures, notes, assignments, and e-resources.

- **Hotel Management System** *CSE 2100 Project*
  Tools: Java, OOP
  ◦ Created a Java-based GUI hotel system with billing, reservations, file handling, and OOP-based modular design.

SKILLS

Programming	C, C++, Java, Python
Web & Databases	HTML, CSS, MySQL
Tools & IDEs	Visual Studio, CodeBlocks, PyCharm, Jupyter Notebook, Google Colab, GitHub, Kaggle, Google Suite, MS Office
Soft Skills	Self-motivated, Multitasking, Detail-Oriented, Adaptability, Time Management, Problem-Solving, Event Management
Interpersonal	Teamwork, Mentoring, Active Listening, Public Speaking, Presentation Skills, Communication

HONORS AND AWARDS

- **National Champion in Hackathon** *2022*
  Moscow Engineering Physics Institute (MEPhI) & Information Center on Nuclear Energy
  ◦ Achieved **1st position** in National Hackatom (a form of the hackathon).

- **Smart Rajshahi Innovation Challenge** *2023*
  S2S Ventures, supported by a2i and Rajshahi City Corporation
  ◦ Selected among the **Top 30** in the Smart Rajshahi Innovation Challenge 2023.

- **Academic Scholarships & Awards** *2017 – 2024*
  Various Institutions
  ◦ **2019–2024:** Government Scholarship in Dhaka Board, HSC.
  ◦ **2019:** DMP (Dhaka Metropolitan Police) Scholarship, HSC.
  ◦ **2017:** Prime Minister's Scholarship, SSC.
  ◦ **2017:** DMP Head Office Scholarship, SSC.
  ◦ **2017:** Excellence Award, SSC Result, BRAC.

Leadership & Development Programs

- **Banglalink Womentor 5.0 Program** *Jul 2024 – Dec 2024*
  Banglalink Head Office, Gulshan-1, Dhaka
  ◦ Selected among the Top 15 Female STEM talents nationwide.
- **Aspire Leaders Program 2024** *Apr 2024 – Jul 2024*
  Organised by the Aspire Institute, founded at Harvard University
  ◦ Selected and participated in an extensive leadership development program.

Professional Training

- **Python (Django) Training** – Organized by RUET under EDGE Project, BCC, ICT Division *May 2024 – Feb 2025*
- **Python Training (PCEP Certified, 93% Score)** – PeopleNTech Institute of IT & ICT Division *Jun 2023 – Nov 2023*
- **LEDP Training (Web Design & Development)** – Bangladesh Government's LEDP Program *Jan 2023 – Apr 2023*

Extracurricular Activities

◦ **Participant**, International Astronomical Search Colaboration (Dec 2020 – Jan 2021).
◦ **Community Member**, DNS Dhaka Nibash Somiti at RUET (Jan 2020 – Jul 2025)..
◦ **Volunteer**, Decoration Team, CSE Fest, RUET (2023).
◦ **Volunteer**, RUET Admission Process (2021).
◦ **Member**, Decoration Club at HCC (2017–2019).
◦ **Logistics Volunteer**, Science Fest, Holy Cross College (2018).
◦ **Campus Ambassador** for KUET BitFest (2024).

Language Proficiency

Bangla	Native proficiency (Speaking, Writing, Listening)
English	Professional proficiency (Speaking, Writing, Listening)

References

1. **Shyla Afroge**
   Associate Professor, Department of CSE
   Rajshahi University Of Engineering & Technology (RUET)
   Email: shyla.ruet@gmail.com
2. **Md. Azmain Yakin Srizon**
   Assistant Professor, Department of CSE
   Rajshahi University Of Engineering & Technology (RUET)
   Email: azmainsrizon@gmail.com